A Guide to Autoregressive Distributed Lag Models for Impulse Response Estimations*

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Abstract

We provide a guide to using autoregressive distributed lag models for impulse response estimations with an identified structural shock or an external instrument for the shock. We illustrate how specifications widely used in practice can lead to inconsistent and inefficient estimators. We further review empirical results from previous papers and show that some results appear to be statistical artifacts. We propose a simple method to avoid such a false conclusion from biases or large standard errors and obtain consistent and precise estimates.

Keywords: autoregressive distributed lag models, single equation regression, impulse responses, model selection

JEL Codes: C13, C29, C52, G01.

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I Introduction

Understanding the dynamic responses of aggregate variables to structural shocks has been a central question in macroeconomics. While many researchers employ traditional vector autoregressive (VAR) models to estimate responses, single equation methods are becoming popular alternatives in applied macroeconomic research. These methods are simple, flexible, and easy to understand. Moreover, these methods allow researchers to concentrate on a single shock and a single endogenous variable because single equation methods do not require full specifications of other shocks.

This paper studies a specific single equation method that estimates structural impulse responses via an autoregressive model with distributed lags of an identified shock or an external instrument for the shock. We use autoregressive distributed lag (ADL) models as a tool to estimate the impulse response coefficients rather than describe the underlying data generating process (DGP) in every respect. Therefore, our focus differs from that of the classic literature concerning distributed lag models mainly in the search for the identification of "true" models (see Hendry, Pagan and Sargan, 1984; Sims, 1973, and references therein). Instead, we investigate the properties of the estimated structural impulse response function (IRF) based on ADL models when an identified shock or an external instrument is available.

ADL models, often referred to as single equation regressions following Romer and Romer (2004, 2010), have a long history and have been actively used until recently (e.g., Alesina, Favero and Giavazzi, 2015; Arezki, Ramey and Sheng, 2017; Cerra and Saxena, 2008; Coibion, 2012; Romer and Romer, 2004, 2010). However, knowledge concerning their properties in terms of the derived impulse responses is limited, and guidance for practitioners is lacking. We investigate the properties of ADL models and provide a simple recipe to tailor these models according to various research questions. Our findings could help researchers better understand their estimators and estimate impulse responses consistently.

Specifically, we focus on the following three parameters: the length of the distributed lags of the identified shocks or the external instruments (denoted by A), the order of autoregression (B), and the longest horizon of the impulse responses relevant to the research questions (C). We examine how A, B, and C affect the estimators of IRFs. We further provide a practitioner's guide to selecting A and B in a manner that guarantees consistency of the estimator up to horizon C. Our result holds in a generic setup regardless of the true ADL representation of the data generating process (DGP). When researchers do not know the true

lag orders, which is usually the case because macroeconomic variables are driven by multiple structural shocks, our guide can provide a reliable method to consistently estimate impulse responses using ADL models. Furthermore, when a structural shock is not observable, our procedure can point-identify impulse responses up to normalization using an external instrument. Our findings echo similar results reported in Stock and Watson (2018) regarding the method of external instruments for structural VARs and local projections (LP).

We show that we can consistently estimate impulse response coefficients up to horizon C by making A greater than or equal to C. This result holds for a generic setup, including cases in which the true lag orders are infinite. However, it is not unusual for A to be much smaller than C in applied research. We explain why, in this case, it is easy to falsely conclude that a shock has long-run effects on a dependent variable with no relation to the true impulse responses. Given that consistency is achieved by long distributed lags ($A \geq C$), B can further affect the variability of the estimates; the estimator can become more efficient by properly choosing B. Therefore, given C, which naturally follows from the research question, we suggest setting A at C and relying on standard order selection criteria for autoregressive models to choose B. Depending on DGPs, our consistency results may be accompanied by an increase in variance of the impulse response estimator, especially at long horizons. Thus, regarding the bias-variance trade-off, setting $A \geq C$ prioritizes eliminating (asymptotic) biases in a generic setup.

Our suggestion differs from the approaches employed in previous papers. Without an established procedure, researchers have relied on statistical tests (Cerra and Saxena, 2008), used an information criterion to jointly choose A and B (Coibion, 2012), or selected a number that might be justified by reasonable results (Arezki, Ramey and Sheng, 2017; Hsiang and Jina, 2014; Kilian, 2008; Romer and Romer, 2004, 2010). All these methods are potentially problematic because the selected A can be much smaller than C. When $A < C$, the estimates of the impulse response at horizons between A and C tend to flatten regardless of the true responses. As a result, a researcher may erroneously find statistically significant long-run effects. For example, when $A < C$, it may appear that a financial crisis has persistent negative effects on economic growth and that the economy does not recover from the crisis permanently. However, the statistical significance of the estimates at horizons between A and C may disappear when $A = C$ (Section III). Furthermore, an arbitrarily selected B renders the estimator more variable, widens the confidence intervals, and lowers the power of statistical tests. For instance, one may not detect the negative responses of unemploy-

ment rates to expansionary military news shocks because of large standard errors (Section III). Therefore, understanding the properties of ADL models is important for both guiding applied macroeconomic research and correctly interpreting the results of previous studies.

Related Literature. The method employed in this paper is based on identified structural shocks or external instruments for such shocks. Although constructing a series of structural variations may appear daunting, substantial progress has been achieved in this direction, and thus far, numerous series have been credibly identified (see Nakamura and Steinsson, 2018*b*; Ramey, 2016, for a review).[1]

After securing identified structural innovations or instruments for these innovations, practitioners can use ADL models, LPs à la Jordà (2005), or bivariate (or small-scale) VARs to estimate impulse response functions. In this paper, we focus on ADL models and provide guidance to practitioners. Knowledge regarding the estimated impulse responses in this popular setup with observable structural shocks (or external instruments for the shocks) is limited, although this approach is widely employed in applied research (e.g., Alesina, Favero and Giavazzi, 2015; Arezki, Ramey and Sheng, 2017; Cerra and Saxena, 2008; Coibion, 2012; Hsiang and Jina, 2014; Romer and Romer, 2004, 2010).

LPs are an increasingly popular alternative to ADL models (Chang and Sakata, 2007; Jordà, 2005; Stock and Watson, 2018). Although an extensive comparison between the two methods is beyond the scope of this paper (see Plagborg-Møller and Wolf, 2021, for a comparison between LP and VAR), we believe that ADL models still represent a useful and relevant method for the following reasons. First, LPs tend to generate erratic and spiky impulse response functions (see Barnichon and Brownlees, 2019; Ramey, 2016). Moreover, Gorodnichenko and Lee (2020) report that with practically relevant sample sizes, LPs underestimate responses when the true effects are persistent (see also Herbst and Johannsen, 2020, for bias in LPs). Our simulations show that ADL models deliver results that are more robust to these concerns than LPs (Online Appendix G). Finally, ADL models allow for

[1] As a few examples, measures of productivity shocks are provided in Alexopoulos (2011), Basu, Fernald and Kimball (2006), and Fernald (2014); monetary policy innovations are provided in Gürkaynak, Sack and Swanson (2005), Kuttner (2001), Nakamura and Steinsson (2018*a*), and Romer and Romer (1989, 2004); unanticipated variation in fiscal policy is discussed in Auerbach and Gorodnichenko (2012), Mertens and Ravn (2013), Miyamoto, Nguyen and Sergeyev (2018), Ramey (2011), Ramey and Zubairy (2018), and Romer and Romer (2010); oil shocks in Arezki, Ramey and Sheng (2017) and Kilian (2008); financial crises and political events are discussed in Acemoglu et al. (2019), Cerra and Saxena (2008), Jalil (2015), and Romer and Romer (2017); and unpredictable natural disasters are discussed in Hsiang and Jina (2014).

simple and convenient tests of joint hypotheses regarding impulse response coefficients at different horizons. For LPs, however, we need to estimate a set of multiple equations to obtain impulse response coefficients at different horizons. Therefore, the number of parameters in the system of equations can easily be too large to reliably estimate the variance-covariance matrix of the estimator to test joint hypotheses.

Another method widely used in practice is bivariate VARs with a shock and an endogenous variable (e.g., Alexopoulos, 2011; Barakchian and Crowe, 2013; Basu, Fernald and Kimball, 2006; Jalil, 2015; Romer and Romer, 2004). We find that in this specific case, ADL models can be more efficient and less biased than bivariate VARs (Online Appendices D and E). Therefore, ADL models can be a useful tool in applied macroeconomic research, and the properties of this method are worth investigating. Here, we do not intend to argue that ADL models are always better than any VAR model. Multivariate approaches have merits, such as utilizing more information, capturing the transmission mechanism of a shock through endogenous variables, and their close relation to dynamic stochastic general equilibrium (DSGE) models. To maintain a focused discussion, we concentrate on ADL models in this paper. A formal comparison of ADL and various VAR models remains a topic for future research.

There is a large body of classic literature concerning distributed lag models (see Hendry, Pagan and Sargan, 1984; Sims, 1973, and references therein). The studies in this literature implicitly assume that a "true" model yielding a white-noise error exists. Then, these studies search for the correct specification of the DGP with or without additional restrictions on lag weights.[2] Our objective differs because we focus on the impulse response coefficients rather than the fit of the ADL model. In this paper, ADL models serve only the purpose of estimating impulse response coefficients rather than describing or approximating the whole true DGP in every respect. For example, our consistency results of the estimated IRFs do not require the selected model with finite A and B to represent the true DGP. Focusing directly on the performance of the estimated impulse response coefficients is necessary because two ADL models of similar quality in terms of the least squares fit can deliver substantially different derived statistics, including the impulse response coefficients (Hendry, Pagan and Sargan, 1984; Sims, 1972).

Our approach requires "long" distributed lags to secure consistency of the impulse re-

[2]These restrictions include the low order polynomial distributed lag (Almon, 1965), the rational distributed lag Jorgenson (1966), and the "smoothness priors" described by Shiller (1973).

sponse estimators because the length of the distributed lags (A) should be greater than or equal to the longest horizon relevant to the research questions (C). A related but different approach in the literature is to adopt a long autoregressive model with a large B to approximate the population moving-average process (Bühlmann, 1995; Chang and Sakata, 2007; Kuersteiner, 2005; Poskitt, 1994, 2007). The major goal of these papers is to recover moving-average coefficients associated with Wold innovations. However, such Wold innovations do not need to coincide with structural shocks when the DGP is noninvertible. As a result, impulse responses to Wold innovations do not need to have a structural interpretation in which macroeconomists are interested (see Hansen and Sargent, 1980, 1991; Fernández-Villaverde et al., 2007; Lippi and Reichlin, 1993; Forni, Gambetti and Sala, 2019; Wolf, 2020). A "solution" to this problem is to seek external variations that have direct structural meanings (see Kilian and Lütkepohl, 2017, Ch. 17 for a review).[3] Our contribution is aimed at helping researchers who adopt this approach to the identification problem. We provide a simple guide to consistently and efficiently estimating structural impulse response functions. Furthermore, our guide does not require prior knowledge of the true lag structure of the DGP.

Outline. The remainder of the paper is organized as follows. Section II presents the statistical properties of IRFs from ADL models in detail with a focus on consistency and efficiency. Section III provides applications of our results to real-world data. We examine the long-run effects of financial and political crises on economic growth as described in Cerra and Saxena (2008) and highlight the relationship between consistency and the choice of A given C. To illustrate the effects of B on the variability of the estimator, we rely on the military news shocks constructed by Ramey and Zubairy (2018) and unemployment rates. This example is provided for expositional purposes and is not directly related to the analyses in Ramey and Zubairy. Section IV concludes this paper.

II Properties of Impulse Responses from ADL Models

We consider a generic setup of ADL models. Let x_t be a structural shock and y_t denote an endogenous variable of interest. We assume that the dynamics of y_t consist of two parts,

[3] Another method is to consider a long vector autoregression of a shock and an endogenous variable, which will be compared with ADL models in Online Appendix D.

namely, one part related to x_t and the remainder, as follows:

$$y_t = \sum_{c=0}^{\infty} \psi_{x,c} x_{t-c} + u_t = \psi_x(L) x_t + u_t, \tag{1}$$

where $\psi_x(L)$ represents the IRF that we intend to estimate, and u_t captures a component driven by all shocks other than x.

A few standard assumptions are made regarding Equation (1) as follows. (i) $\{x_t\}$ follows a white-noise process with standard deviation σ_x. (ii) Δu_t admits a $MA(\infty)$ representation $\Delta u_t = g_y + \sum_{c=0}^{\infty} \psi_{e,c} e_{t-c} = g_y + \psi_e(L) e_t$, where $\{e_t\}$ is a white-noise process with standard deviation σ_e. (iii) $E[x_{t+r} e_{t+s}] = 0$ for all r and s. (iv) $\sum_{c=0}^{\infty} \psi_{x,c}^2 < \infty$ and $\sum_{c=0}^{\infty} \psi_{e,c}^2 < \infty$. (v) $\psi_{e,0} = 1$. For (ii) and (iv), one can easily extend our results to cases in which either x_t has a permanent effect on y_t or u_t is stationary without first-difference. We impose the stationarity of $\{\psi_x(L) x_t\}$ and $\{\Delta u_t\}$ in consideration of a monetary policy shock and real gross domestic product (GDP) as examples of x_t and y_t.

In practice, structural shock x_t may not be directly observable. Empirically constructed shocks may involve measurement errors and reflect only a part of the true shock. Therefore, to enhance the applicability of our results, we assume that an external instrument z_t for the structural shock x_t is observable. We consider cases with a scalar instrument to maintain a focused discussion. When multiple external instruments are available, although rare in practice, any linear combination of these instruments can be used as z_t in the following discussion.

Similar to Stock and Watson (2018), we make the following standard assumptions regarding z_t. (vi) $\{z_t\}$ is a serially uncorrelated process with mean zero and standard deviation $\sigma_z > 0$.[4] (vii) (relevance) $Corr(z_t, x_t) = \rho \neq 0$. (viii) (contemporaneous exogeneity) $Cov(z_t, e_t) = 0$. (ix) (lead-lag exogeneity) $Cov(z_t, x_{t+l}) = Cov(z_t, e_{t+l}) = 0$ for all $l \neq 0$. In a special case where $\rho = 1$, structural shock x_t is observable through z_t up to scale.

To estimate the impulse response coefficients, we run an ordinary least squares (OLS)

[4]For a serially correlated series, we use "shock" components after purging out autoregressive dynamics from the original measure. Consider, for illustration, the utilization-adjusted TFP series constructed by Fernald (2014). As in Ramey (2016), we can relate z_t to the first difference of the TFP series, removing the serial correlations and time-trends in the level of TFP. When necessary, the standard errors of the OLS estimator $\hat{\eta}$ of η in Equation (2) need to be adjusted in accordance with the generated regressors $z_t, \ldots, z_{t-A}$ (see, e.g., Pagan, 1984).

regression of the following ADL model:

$$\Delta y_t = \delta + \sum_{a=0}^{A} \alpha_a z_{t-a} + \sum_{b=1}^{B} \beta_b \Delta y_{t-b} + v_t = \zeta_t' \eta + v_t, \tag{2}$$

where v_t is the projection residual that can be serially correlated due to truncation of the lag series, $\zeta_t = (1, z_t, z_{t-1}, \ldots, z_{t-A}, \Delta y_{t-1}, \Delta y_{t-2}, \ldots, \Delta y_{t-B})'$, and $\eta = (\delta, \alpha_0, \ldots, \alpha_A, \beta_1, \ldots, \beta_B)'$. Let φ_c be $\psi_{x,c} - \psi_{x,c-1}$ for all $c \geq 0$, where $\psi_{x,-1} \equiv 0$. Given the OLS estimate $\hat{\eta}_t = \left(\hat{\delta}, \hat{\alpha}_0, \ldots, \hat{\alpha}_A, \hat{\beta}_1, \ldots, \hat{\beta}_B\right)' = \left(\sum_t \zeta_t \zeta_t'\right)^{-1} \left(\sum_t \zeta_t \Delta y_t\right)$, the following equations deliver the estimated IRF:

$$\begin{aligned} \hat{\varphi}(L) &= \hat{\varphi}_0 + \hat{\varphi}_1 L + \ldots = \left(1 - \hat{\beta}_1 L - \ldots - \hat{\beta}_B L^B\right)^{-1} \left(\hat{\alpha}_0 + \ldots + \hat{\alpha}_A L^A\right), \\ \hat{\psi}_x(L) &= \hat{\psi}_{x,0} + \ldots + \hat{\psi}_{x,C} L^C + \ldots, \quad \text{where} \quad \hat{\psi}_{x,c} = \hat{\varphi}_0 + \ldots + \hat{\varphi}_c \quad \text{for all} \quad c \geq 0. \end{aligned} \tag{3}$$

Our objective is to study the properties of $\hat{\psi}_{x,c}$ in Equation (3). Specifically, we investigate the sensitivity of $\hat{\psi}_{x,c}$ to A, B, and C, where A is the length of the distributed lag, B is the order of the autoregressive component, and C is the longest horizon of the impulse response analysis that naturally follows from the research question. Below, we show how these key parameters are related to the consistency and efficiency of the estimator in Equation (3).

Consistency and Distributed Lag

We begin with the relationship between A and the consistency of the estimated impulse response coefficients. For this purpose, we fix A and B then increase the sample size T to infinity to investigate a probability limit of $\hat{\psi}_{x,c}$. We show that the asymptotic behavior of $\hat{\psi}_{x,c}$ depends on whether $c \leq A$ or $c > A$.

Before discussing the asymptotic properties of $\hat{\eta} = \left(\sum_t \zeta_t \zeta_t'\right)^{-1} \left(\sum_t \zeta_t \Delta y_t\right)$ and the IRF estimator $\{\hat{\psi}_{x,c}\}$, we first show that the population projection coefficient of Δy_t on ζ_t, denoted by $\bar{\eta} \equiv \left(E\left[\zeta_t \zeta_t'\right]\right)^{-1} E\left[\zeta_t \Delta y_t\right]$, is well-defined.

Proposition 1. *$E\left[\zeta_t \zeta_t'\right]$ is nonsingular. Therefore, $\bar{\eta} = \left(E\left[\zeta_t \zeta_t'\right]\right)^{-1} E\left[\zeta_t \Delta y_t\right]$ is well-defined.*

Proof. See Online Appendix A1. □

To facilitate the discussion, we assume that the weak law of large numbers holds for $\{\zeta_t \zeta_t'\}$ and $\{\zeta_t \Delta y_t\}$ for fixed A and B.[5] Thus, $\hat{\eta} = (\hat{\delta}, \hat{\alpha}_0, \ldots, \hat{\alpha}_A, \hat{\beta}_1, \ldots, \hat{\beta}_B)'$ converges in

[5] See Online Appendix A1 for a set of sufficient conditions.

probability to $\bar{\eta}$ as T increases to infinity. Then, we compare $\hat{\psi}_{x,c}$ based on $\hat{\eta}$ as shown in Equation (3) with $\psi_{x,c}$. The following is our first main result.

Proposition 2. *Fix A and B. As $T \to \infty$, $\hat{\psi}_{x,c}$ converges in probability to $\theta\psi_{x,c}$ for any $c \leq A$, where $\theta = \frac{\rho\sigma_x}{\sigma_z}$.*

Proof. See Appendix. □

In a special case in which x_t is directly observable (i.e., $z_t = x_t$), $\theta = \frac{\rho\sigma_x}{\sigma_z} = 1$. Then, Proposition 2 yields the consistency of the estimated IRF $\{\hat{\psi}_{x,c}\}$ up to horizon A.

Corollary 1. *If $\theta = 1$, which is true when $x_t = z_t$, $\hat{\psi}_{x,c}$ converges in probability to $\psi_{x,c}$ for any $c \leq A$ as $T \to \infty$.*

In some applications, structural shock x_t may not be observable. Even in this case, we can still identify the "shape" of the IRF up to horizon A because θ in the probability limit of $\hat{\psi}_{x,c}$ is common to all $c \leq A$, and $\theta \neq 0$. Thus, an external instrument z_t in the ADL model (2) allows us to consistently estimate the *relative* IRF.

Corollary 2. *For any $c_1, c_2 \leq A$ such that $\psi_{x,c_1} \neq 0$, $\frac{\hat{\psi}_{x,c_2}}{\hat{\psi}_{x,c_1}}$ converges in probability to $\frac{\theta\psi_{x,c_2}}{\theta\psi_{x,c_1}} = \frac{\psi_{x,c_2}}{\psi_{x,c_1}}$ as $T \to \infty$.*

Remark 1. Therefore, in a generic setup, we can consistently estimate the (relative) IRF up to horizon A that is the length of the distributed lag. Consider a case where x_t is directly observable. It is not surprising per se that a distributed lag regression ($B = 0$) of Δy_t on the lagged values of x_t yields a consistently estimated IRF for $c \leq A$ (see, e.g., Romer and Romer, 2010, Equation (6)). Our main result substantially generalizes this simple case in two respects: First, controlling for the lagged endogenous variable by including an arbitrary number B of lags of Δy_t does not destroy the consistency of the IRF up to horizon A. Second, our results hold for the models with an external instrument, which is not necessarily identical to a structural shock ($z_t \neq x_t$). Autoregressive parts ($B > 0$) can improve precision in the estimated IRF by purging out predictable dynamics, and allowing for external instruments significantly increase the applicability of our approach. Furthermore, Proposition 2 and the following corollaries hold for any A and B regardless of their "true" values. Even when the ADL model (2) with the selected A and B is not "correctly" specified and therefore is not a "good" approximation of the true DGP in every respect, we can still consistently estimate the (relative) impulse response coefficients up to horizon A. Thus, our method can reduce

the burden of finding the "true" A and B. This result holds even when the true lag orders are infinite.

Remark 2. When structural shock x_t is not observable, it is not possible to separately identify $\{\psi_{x,c}\}$ and σ_x^2. To resolve this scale ambiguity, Stock and Watson (2018) propose a normalization by $\psi_{x,0}$ without loss of generality. Stock and Watson further discuss the identification of the relative IRF $\{\frac{\psi_{x,c}}{\psi_{x,0}}\}$ in structural VARs and LPs. Our result implies that a similar identification based on an external instrument can be achieved in ADL models because the probability limits of $\hat{\psi}_{x,c}$ share the same scale factor of $\theta \neq 0$.

Remark 3. θ is a regression coefficient of x_t on z_t. Therefore, $x_t = \theta z_t + \omega_t$, where $\{\omega_t\}$ is a white-noise process that is orthogonal to $\{z_t\}$ and $\{e_t\}$. Then, Equation (1) yields $\Delta y_t = g_y + \varphi(L)x_t + \psi_e(L)e_t = g_y + \varphi(L)\theta z_t + [\varphi(L)\omega_t + \psi_e(L)e_t]$. By relying on the Wold decomposition, we can redefine the terms inside the square brackets as $\tilde{\psi}_e(L)\tilde{e}_t$ for lag series $\tilde{\psi}_e(L)$ and white-noise process $\{\tilde{e}_t\}$. Thus, we can consider z_t a "shock" in the following representation of the DGP: $\Delta y_t = g_y + \varphi(L)\theta z_t + \tilde{\psi}_e(L)\tilde{e}_t$. Because $\varphi(L) = (1-L)\psi_x(L)$, the corresponding IRF $\{\frac{\partial y_t}{\partial z_{t-c}}\}$ is given by $\{\theta\psi_{x,c}\}$. Then, Proposition 2 implies that the IRF estimators using z_t in Equations (2) and (3) "consistently" estimate the corresponding IRF $\{\theta\psi_{x,c}\}$ for c being less than or equal to the length of the distributed lag A.

Next, we consider cases in which $c > A$. Below, for exposition, we assume that x_t is directly observable (i.e., $z_t = x_t$). However, similar results apply to the relative impulse responses when z_t does not coincide with x_t. To simplify the notations, let a bar denote the probability limit of an estimator. For example, $\bar{\alpha}_1$ is the probability limit of $\hat{\alpha}_1$. The following proposition (borrowed from Brockwell and Davis, 1991, Equation (3.3.4)) characterizes the estimator at horizons longer than A.

Proposition 3. *$\hat{\varphi}_{A+c}$ is obtained recursively using*

$$\hat{\varphi}_{A+c} = \hat{\beta}_1\hat{\varphi}_{A+c-1} + \ldots + \hat{\beta}_B\hat{\varphi}_{A+c-B}, \tag{4}$$

given the initial condition $(\hat{\varphi}_{A-B+1}, \ldots, \hat{\varphi}_A)'$ for all $c > 0$, where $\hat{\varphi}_d = 0$ for $d < 0$. As $T \to \infty$, $\bar{\varphi}_{A+c}$ is determined by a similar process with $\bar{\beta}_b$'s and the initial condition $(\varphi_{A-B+1}, \ldots, \varphi_A)'$.

This result has a simple but crucial implication.

Remark 4. $\hat{\psi}_{x,c}$ tends to flatten for $c > A$ regardless of the true shape of the IRF. Therefore, $\hat{\psi}_{x,c}$ is an inconsistent estimator for $c > A$ in a generic setup. To illustrate this result, we consider the recursive representation of $\hat{\varphi}_{A+c}$ in Proposition 3. When the lag polynomial $\left(1 - \hat{\beta}_1 L - \ldots - \hat{\beta}_B L^B\right)$ is stable, $\hat{\varphi}_{A+c}$ exponentially decays to zero in c (Brockwell and Davis, 1991, p. 108). Moreover, if this decay is fast, which is the case with real GDP (Section III) and industrial production (Online Appendix E1), $\sum_{c=A+1}^{\infty} \hat{\varphi}_c$ becomes small. Next, note that the estimated long-run effect $\lim_{c\to\infty} \hat{\psi}_{x,c}$ can be written as follows:

$$\lim_{c\to\infty} \hat{\psi}_{x,c} = \sum_{c=0}^{\infty} \hat{\varphi}_c = \hat{\psi}_{x,A} + \sum_{c=A+1}^{\infty} \hat{\varphi}_c. \tag{5}$$

When the last term is small, $\hat{\psi}_{x,c}$ flattens around $\hat{\psi}_{x,A}$ for $c > A$.

This feature may have a substantial effect on inferences regarding the long-run effects of a shock. When A is too small and $\hat{\psi}_{x,A}$ is far from zero, one might mistakenly find persistent effects with no relation to the true effects. We will return to this issue when we discuss the economic (non-)recovery from a crisis in Section III.

Remark 5. $\hat{\psi}_{x,c}$ is a consistent estimator for all $c \geq 0$ if Equation (2) exactly coincides with the true DGP. Furthermore, in some cases, a long autoregression (large B) helps ADL models resolve the issue stated in Remark 4. For example, when $u_t = 0$ and $\varphi(L)$ is invertible, $\varphi(L)^{-1}\Delta y_t = x_t$. As a result, the DGP admits a representation (2) with $A = 0$ and B being the order of $\varphi(L)^{-1}$. In another case, $\psi_e(L)$ is invertible, and $\varphi_A(L) \equiv \psi_e^{-1}(L)\varphi(L)$ is a lag polynomial with order $A < \infty$. By ignoring an irrelevant constant, the DGP (1) becomes $\psi_e(L)^{-1}\Delta y_t = \varphi_A(L)x_t + e_t$. In this case, given a fixed A, increasing B may help us better estimate $\psi_{x,c}$ for $C \geq c > A$.

However, our strategy for consistently estimating the impulse responses is to include a long distributed lag ($A \geq C$) rather than long autoregression or attempting to find the true A and B. We suggest a long distributed lag because information regarding the true lag structure of the DGP is likely limited in practice. Because our approach yields consistently estimated (relative) impulse responses in a generic setup, as shown in Proposition 2, our method can reduce the burden of finding the true A and B or ensuring that the order of $\varphi_A(L)$ is finite.

Monte Carlo Simulations. The results of the Monte Carlo studies illustrate our findings. We simulate a DGP with a hump-shaped IRF $\{\psi_{x,c}\}$ and $\{\Delta u_t\}$ following an AR(2) process,

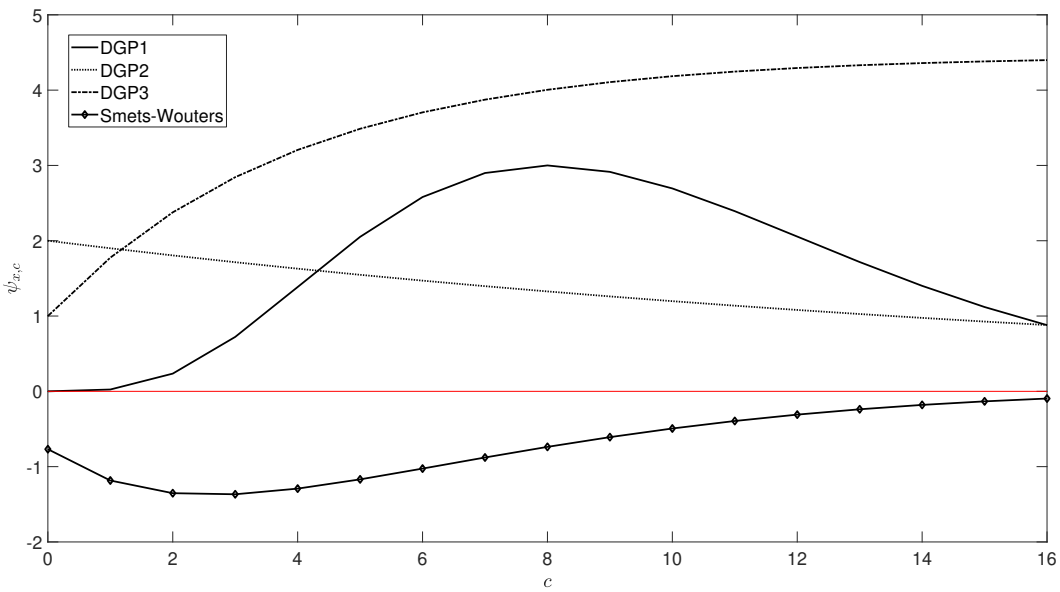

Figure 1: Population Impulse Response Function $\{\psi_{x,c}\}$ for Monte Carlo Simulations

Notes: The graph shows the impulse response function $\{\psi_{x,c}\}$ of the DGPs assumed for the Monte Carlo simulations. The shape of the IRF of DGP1 is motivated by the responses of real GDP to expansionary monetary policy shocks studied by Coibion (2012). Regarding DGP2, we assume that $\psi_{x,c} = 2(0.95)^c$. DGP3 features $\varphi(L) = (1 - 0.78L)^{-1}$ and $\psi_{x,c} = \varphi_0 + \varphi_1 + \ldots + \varphi_c$, yielding a positive long-run response. Finally, we consider the responses of output to contractionary monetary policy shocks in the model of Smets and Wouters (2007).

$\Delta u_t - g_y = \rho_{p1}(\Delta u_{t-1} - g_y) + \rho_{p2}(\Delta u_{t-2} - g_y) + e_t$. x_t and e_t are drawn from independent normal distributions with variances σ_x^2 and σ_e^2, respectively. Finally, we assume that $z_t = x_t$. The solid line in Figure 1 labelled DGP1 depicts the population IRF of this DGP.[6]

For each generated time series of 100 observations, we apply Equations (2) and (3) with $A = 8$ and 16 given $B = 2$.[7] The simulation size is 5,000. The left panel in Figure 2 shows the 5th percentile, the mean, and the 95th percentile of $\hat{\psi}_{x,c}$. Consistent with Proposition 2, the estimates are close to the true values when $c \leq 8$. However, the two specifications with different A remarkably disagree when $c > 8$. In the model with $A = 8$, the estimated impulse response function flattens after A as if there were significant long-run effects, while the true response reverts to zero. Having a large sample does not solve this problem, as implied by Proposition 3 and Equation (5). When the sample size T increases to 500, the

[6]We use the following parameters: $\sigma_x = 1$, $\sigma_e = 1.5$, $g_y = 0.5$, $\rho_{p1} = 0.26$, and $\rho_{p2} = 0.4$. Online Appendix A3 covers other DGPs for u_t. See Online Appendix B2 for a case in which a finite-order AR representation of $\{\Delta u_t\}$ does not exist.

[7]We fix B at 2 because $\{\Delta u_t\}$ follows an AR(2) process. The relationship between B and the $\{\Delta u_t\}$ process is discussed in the following section.

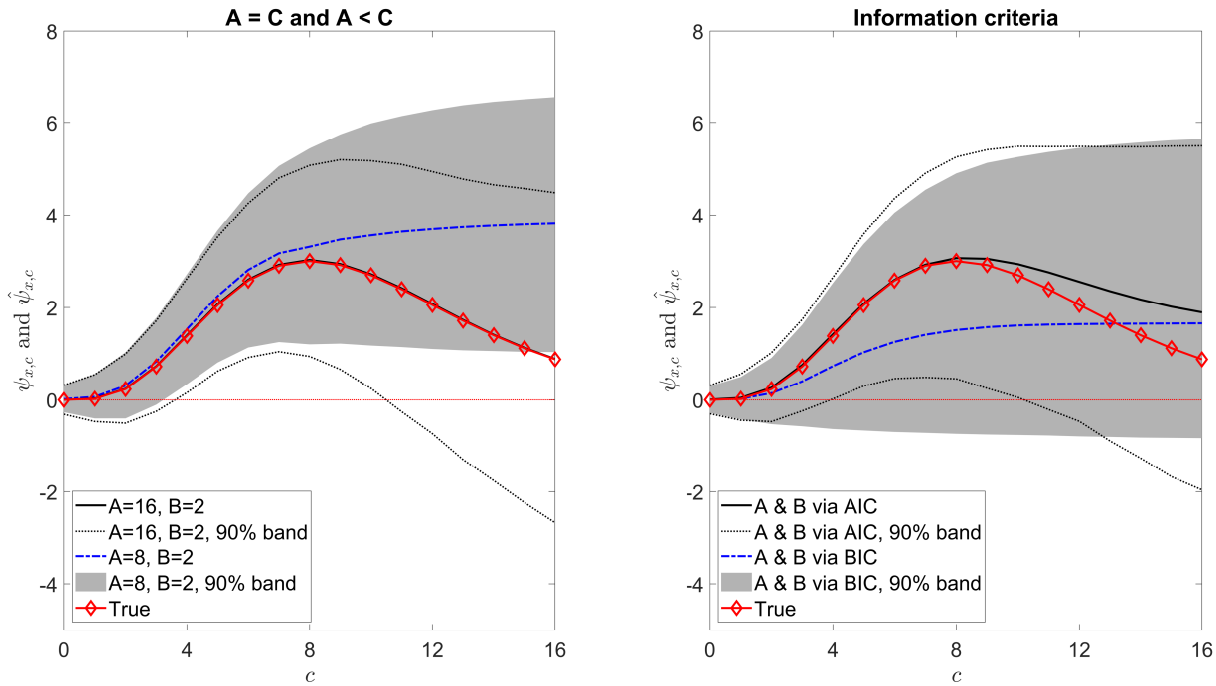

Figure 2: Consistency and A

Notes: The red line with diamonds represents the true impulse responses $\psi_{x,c}$ for $c \leq C = 16$. In the left panel, the black solid line illustrates the average estimates $E[\hat{\psi}_{x,c}]$ when $A = 16$ and B is fixed at the true order 2. The bottom and the top 5th percentiles of the simulated $\hat{\psi}_{x,c}$'s are shown by the dotted lines. The blue dash-dotted line represents the average estimates when $A = 8$ and $B = 2$, and the shaded area includes 90% of the corresponding estimates. The right panel illustrates the results when A and B are jointly selected by information criteria. The black solid and dotted lines represent the mean, the 5th percentile, and the 95th percentile of $\hat{\psi}_{x,c}$ when the AIC is used. The blue dash-dotted line and the shaded area show the results based on the BIC. Fixing B at 2 and using an information criterion only for A does not significantly change our results. The sample and simulation sizes are 100 and 5,000, respectively.

average estimates do not significantly change, while the 90% band becomes narrower.[8] This problem does not occur when $A = C = 16$.

Neither the Bayesian nor the Akaike information criterion delivers satisfactory results when A and B are jointly selected. Regarding the Bayesian information criterion (BIC), the selected value of A is only 2.1 on average. Therefore, the estimated impulse response function starts to flatten too early and misses the hump-shaped pattern in the population (see the right panel in Figure 2). While the Akaike information criterion (AIC) yields a larger A and performs better than the BIC (e.g., the mean value of A is 8.7), it also suffers from a similar problem when $c \geq 10$.

[8]This result is available upon request.

In summary, ADL models provide consistent estimators of the true (relative) impulse response coefficients in a generic setup when the associated horizon is less than or equal to the length of the distributed lag. Through an example, we also illustrate that information criteria may fail to deliver a reliable model when we estimate impulse response coefficients with practical sample sizes. Furthermore, the biases at horizons longer than A may generate misleading results that appear to support the existence of long-run effects. Such biases can be crucial when we discuss important macroeconomic questions, such as recovery from a long-lasting recession due to financial crises, monetary neutrality in the long run, and long-term effects of fiscal austerity measures. Therefore, in such cases, we can ensure consistency of the estimator in Equation (3) by including as many lags of a shock in model (2) as the longest horizon of the impulse response analysis.

In some cases, an A smaller than C may be chosen due to the large number of parameters in ADL models when $A = C$ and, therefore, potentially large variances of the IRF estimator. Even in this case, we propose using a specification with $A = C$ to check the robustness of the results. We believe that this robustness check can provide a useful result, specifically when the shape of the IRF and statistical inferences at long horizons are central to the research question.

Efficiency and Autoregression

In this section, we first illustrate what β_b represents. Based on the result, we show that the selection of B resembles the conventional problem of specifying the lag order of an autoregressive process in practically relevant cases. Notably, there are well-developed standard methods for this problem (see Kilian and Lütkepohl, 2017, Ch. 2.6). While we employ the BIC following Schwarz (1978) for illustrative purposes, there may be other approaches for choosing B with similar results.

For a simple exposition, we assume that a lag series $\psi_e(L)$ is invertible.[9]

Proposition 4. *The DGP in Equation* (1) *admits the following representation:*

$$\Delta y_t = \delta + \alpha(L)x_t + \beta(L)\Delta y_{t-1} + e_t, \tag{6}$$

[9]When this assumption does not hold, we can rely on the Wold decomposition. If $\{\Delta u_t\}$ is a covariance-stationary process without a time-varying deterministic component, one can define $\psi_e(L)$ and e_t such that e_t is an element in a closed space spanned by the history of Δu_t. Thus, we can make the residual in Equation (6) arbitrarily close to e_t with appropriate β_b's.

where $1 - L\beta(L) = \psi_e(L)^{-1}$, $\alpha(L) = \psi_e(L)^{-1}(1-L)\psi_x(L)$, *and* $\delta = \psi_e(1)^{-1}g_y$.

Proof. By multiplying Equation (1) by $\psi_e(L)^{-1}(1-L)$, we obtain the following:

$$\psi_e(L)^{-1}\Delta y_t = \psi_e(1)^{-1}g_y + \psi_e(L)^{-1}(1-L)\psi_x(L)x_t + e_t. \tag{7}$$

The results follow from Equation (7). □

Based on Equations (6) and (7), β_b's can be related to $\psi_e(L)$. Under this representation, the autoregressive part in Equation (6), $\sum_{b=1}^{\infty}\beta_b\Delta y_{t-b}$, reflects the regular dynamics of $\psi_e(L)e_t$, which are orthogonal to $\psi_x(L)x_t$. In contrast, the second term on the right-hand side of Equation (7) represents variations related to x_t, which is captured by the distributed lags in Equation (6), $\sum_{a=0}^{\infty}\alpha_a x_{t-a}$.

Remark 6. As shown in Equation (4), $\hat{\varphi}_{x,c}$ for $c > A$ is driven by $\hat{\beta}_b$'s. Because β_b and its estimate are more tightly related to $\psi_e(L)$ than $\psi_x(L)$, $\hat{\varphi}_{x,c}$ and, therefore, $\hat{\psi}_{x,c}$ for $c > A$ do not need to be close to their true values in general. In the simulations, we can induce a remarkably different $\hat{\psi}_{x,c}$ for $c > A$ by assuming other processes for $\{\psi_e(L)e_t\}$ while maintaining the same $\psi_x(L)$ (see Online Appendix A4).

In practice, the longest horizon of impulse response analysis C is usually longer than several years. Given $A \geq C$ in light of the discussion in the previous section, substantial variation in $\psi_e(L)^{-1}(1-L)\psi_x(L)x_t$ at business cycle frequencies is absorbed by the distributed lag of x_t in Equation (2). Therefore, conditioning on $A \geq C$, the choice of B can be considered finding a proper autoregressive representation of $\{\psi_e(L)e_t\}$. We aim to remove predictable variations in this process and leave only a white-noise residual e_t in the place of v_t in Equation (2).[10] However, increasing the order of the autoregressive polynomial decreases the degrees of freedom and increases the variability of the estimator. How to choose a good lag length given the trade-off between purified residuals and high degrees of freedom is not a new question. We follow the conventional approach for an autoregressive model and choose B based on an information criterion following Hurvich and Tsai (1989), Kashyap (1982), and Shibata (1976).

Remark 7. However, one should bear in mind that we use a conventional approach, such as the BIC, only to select B but not A. Applying any information criterion to A might induce

[10] When $\{v_t\}$ is a white-noise process uncorrelated with the regressors in Equation (2), the OLS is an asymptotically efficient estimator of η (see Sims, 1973).

a distributed lag that is too short as shown in Section II. In our view, it is better to avoid overly short distributed lag components as discussed in the previous section unless there are reasons to believe that the order of $\psi_e^{-1}(L)(1-L)\psi_x(L)$ in Equation (7) is low. Instead, one can consistently estimate $\psi_{x,c}$ without having prior knowledge of the lag structure of the DGP by increasing A to C, where C naturally arises from the research question. For example, when the research questions focus on the peak effect of a monetary policy shock, it could be better for C to be greater than 2 years. To study the long-run effects of a crisis, C needs to be at least 5 years.

Monte Carlo Simulation. To assess how B affects the performances of $\hat{\psi}_{x,c}$ in a realistic setup, we simulate the same DGP1 in the previous section. We calculate the mean squared error (MSE) of $\hat{\psi}_{x,c}$ for several B's and the B selected by the BIC for each sample.[11] In these specifications, we fix A at 16. While we employ the BIC to manage the total number of parameters in the regression (2) given a large $A(=C)$, alternative methods may also be used. Similar to the previous simulation, we further include cases in where both A and B are jointly selected by either the BIC or the AIC.

We calculate the MSEs of each estimator and normalize them by the MSE of the estimator when $B=2$ and $A=16$.[12] $B=2$ in this benchmark case because $\{\Delta u_t\}$ follows an AR(2) process and therefore the true order of $\psi_e(L)^{-1}$ is two.[13] The solid line with circles in both panels in Figure 3 represents the relative MSE of $\hat{\psi}_{x,c}$ for each c when B is selected by the BIC and $A=16$, and the other lines illustrate similar measures using other selection rules. The sample size, T, is 100. Clearly, the relative efficiency of the estimators substantially varies with B. Because both overfitting ($B>2$) and underfitting ($B<2$) increase the MSE, an arbitrarily selected B could significantly deteriorate the quality of the estimator. We further find that the BIC works well in determining B. The relative MSEs of the estimator based on the BIC, which are depicted by the line with circles, are small over all horizons. The average value of the selected B is 1.9 (2.0) when the sample size is 100 (500), which is close to the true order. Finally, as illustrated in the right panel, the joint selection of A

[11] The BIC for Equation (2) is given by $BIC(A,B)=\log\left(\sum_{t=l^*+1}^{T}\frac{\hat{v}_t^2}{T-l^*}\right)+(A+B+1)\frac{\log(T-l^*)}{T-l^*}$, where $\hat{v}_t$ is the residual, and l^* is the maximum lag length considered. The first term measures the goodness of fit while the second term is the "penalty" for adding more parameters.

[12] For the decomposition of the MSEs into squared biases and variances and the case in which $T=500$, see Online Appendix B1.

[13] We find similar results in cases in which a finite-order representation of $\psi_e(L)^{-1}$ does not exist (Online Appendix B2).

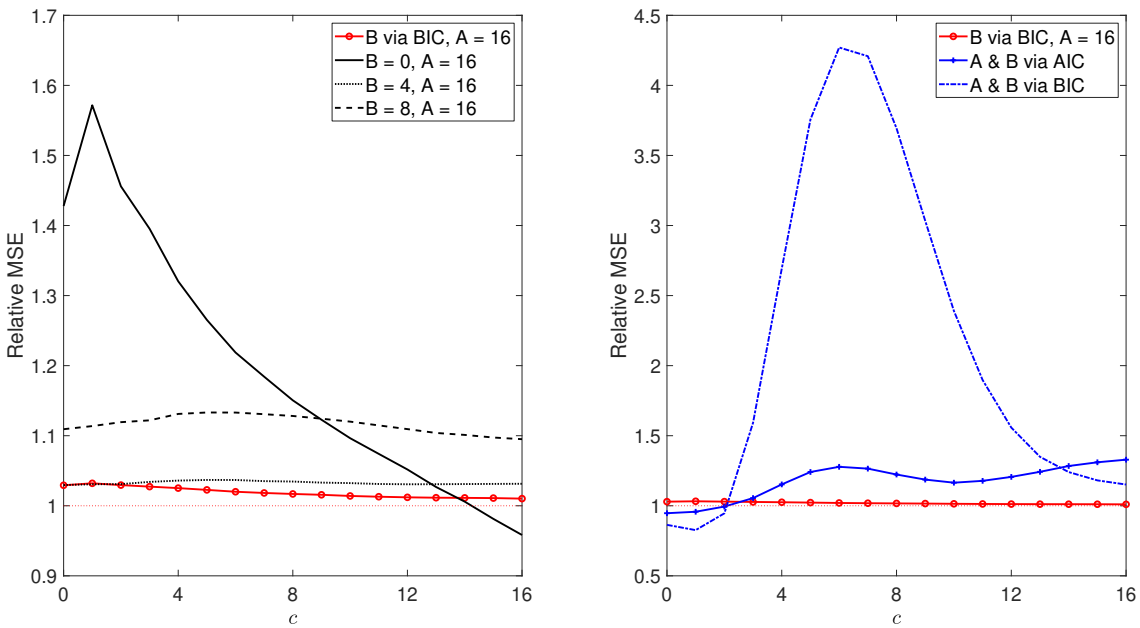

Figure 3: Efficiency and B

Notes: This figure shows the MSE of $\hat{\psi}_{x,c}$ for different B's (and A's) when the sample size is 100. All MSEs are normalized by the MSE of the estimator when $B = 2$ (the true lag order) and $A = C = 16$. The red solid line with circles in both panels represents the (normalized) MSEs of the cases in which B is selected by the BIC. The black solid, dotted, and dashed lines in the left panel represent the results when B equals zero, four, and eight, respectively. In these cases, A is fixed at 16. The blue solid line with plus signs (dash-dotted line) in the right panel illustrates the relative MSE when both A and B are chosen by the AIC (BIC). The simulation size is 5,000.

and B using information criteria generates a substantially larger MSE than our benchmark specification (i.e., $A = C$, B via BIC). This result is true especially at all horizons longer than 2, where meaningful dynamics in y due to x occur.

In summary, the length of autoregressive polynomials affects the efficiency of ADL models for impulse response estimations. In our simulation, the IRF estimator based on our benchmark specification performs reasonably well in terms of the MSEs while successfully capturing the hump-shaped pattern of the IRF over all horizons relevant to the research question.

Lessons from More Monte Carlo Simulations

The illustrative simulation results presented in the previous sections are based on a single DGP with a hump-shaped IRF. To supplement the previous discussion, we carry out more

simulations for various DGPs with different shapes of IRFs (see Figure 1) and true lag orders in the population ADL representation (6). We investigate how the performance of the ADL impulse response estimator $\hat{\psi}_{x,c}$ depends on the selection procedure of A and B in the simulations. We compare our benchmark approach with two other approaches: selecting A and B to minimize the mean integrated squared errors (MISE), $\frac{1}{C+1}\sum_{c=0}^{C} E[\hat{\psi}_{x,c} - \psi_{x,c}]^2$, and jointly choosing A and B by either the BIC or AIC. This section summarizes our simulation results. The full description of DGPs and details of the simulation results (biases, variances, and MISEs) are relegated to Online Appendix C.

Based on extensive simulation results featuring various shapes of the true IRFs and lag orders, we draw three conclusions. First, our benchmark approach with $A = C$ generates small biases at all horizons. This result is robust to different DGPs, confirming the consistency result in a generic setup. Second, for some DGPs, our choice of $A \geq C$ may induce larger variances and MISEs than more parsimonious models. Third, a good ADL model based on information criteria or MISEs does not necessarily yield a good IRF estimator in economic sense. Information criteria tend to choose A that is much smaller than C. A small A may generate substantial biases especially at long horizons by flattening the IRF estimates. A similar pattern arises for MISE-minimizing lag orders.

In summary, our approach with $A = C$ provides consistent estimates at all horizons for all DGPs. This is, however, at the cost of (potentially) larger variances. While our approach with long distributed lags correctly detects the shape of IRFs, it may increase the variability of the estimates. However, even in such cases, our approach generates consistent IRF estimators that perform reasonably well in terms of efficiency in our simulations (see Online Appendix C). Furthermore, when the true lag orders are unknown in practice, the cost of a too short distributed lag can be substantial because of the potential biases in the IRF estimates and misguided economic interpretations.[14]

Taking Stock: A recipe for ADL Models for IRF Estimations

Summarizing our discussion thus far, we propose a simple recipe for selecting A and B given C in ADL models for impulse response estimations as follows:

[14] A researcher may seek to choose a parsimonious model with small A and B as the benchmark specification to reduce the number of parameters in the model. In these cases, checking the sensitivity of the benchmark results to models based on our recipe can be useful for ensuring the robustness of the estimated IRF's shapes and statistical inferences at horizons longer than the benchmark A.

i. Construct/Find a well-identified instrument for the structural shock of interest.

ii. Set A equal to the longest horizon of the impulse response analysis: $A = C$.

iii. Given $A = C$, choose B using a standard model selection method, e.g., an information criterion.

iv. Run the OLS regression of Equation (2) and construct the (relative) impulse responses using Equation (3).

This recipe guarantees consistency of the (relative) impulse responses up to horizon C. Because this result holds regardless of the true orders of the autoregressive and distributed lag components in Equation (6), our procedure can be useful for researchers who do not have strong prior beliefs regarding the lag structure of the DGP as usually is the case in practice when using real-world data. Furthermore, a long distributed lag of instruments can reduce potential (asymptotic) bias, especially for the long-run effects. Finally, given $A = C$, selecting a proper B can improve the power of statistical inferences by reducing the standard error of $\hat{\psi}_{x,c}$.

III Applications

This section presents two applications of ADL models for impulse response estimations: long-run effects of a crisis on economic growth and effects of a military news shock on unemployment rates. We first reassess the validity of the conclusion reported in Cerra and Saxena (2008) that the large output loss from a crisis is highly persistent. Our assessment is based on the relationship between A and the consistency of $\hat{\psi}_{x,c}$ as discussed in Section II. Then, we investigate the responses of unemployment rates to the military news shock constructed by Ramey and Zubairy (2018) and illustrate our points about B and efficiency.

Consistency and Distributed lag

Here, we highlight the importance of A in Equation (2) by focusing on the effects of a crisis on real GDP. Specifically, we examine whether an economy eventually fully recovers from a crisis or whether a crisis results in permanent damage. We use the panel data constructed by Cerra and Saxena (2008, henceforth C&S), which include real GDP and dummy variables for several types of crises.

We estimate a panel version of Equation (2) as follows:

$$\Delta y_{l,t} = \delta_l + \lambda_t + \sum_{a=0}^{A} \alpha_a x_{l,t-a} + \sum_{b=1}^{B} \beta_b \Delta y_{l,t-b} + v_{l,t}, \tag{8}$$

where δ_l and λ_t are country and time fixed effects.[15] The dependent variable is the growth rates of real GDP in country l and year t. The shock variable is a dummy variable for either twin financial or political crises, where the twin financial crises denote cases in which a currency crisis and a banking crisis occur in the same year; the twin political crises imply a similar situation with stronger executive power and civil wars in the same year (see C&S for detail). We choose these two types of crises specifically for illustration as they generate salient differences in the impulse responses depending on A.[16]

The left panels in Figure 4 replicate the results in C&S where $A = B = 4$. The top (bottom) panels illustrate the responses of real GDP to twin financial (political) crises. We compute Driscoll and Kraay (1998) robust standard errors, and the 90 percent confidence bands are shown by the dotted lines.[17] When $A = B = 4$, both twin financial and political crises have a persistent negative effect on economic growth, and the estimates are statistically significant. Thus, the economy appears to not fully recover from a crisis permanently. However, as discussed in Section II, a small A might lead to biased $\hat{\psi}_{x,c}$'s for $c > A$ as $\hat{\psi}_{x,c}$ tends to flatten regardless of the true shape. Thus, the right panels are based on our preferred specification with $A = C = 8$ and $B = 1$. For B, we rely on sequential tests following Han, Phillips and Sul (2017) because standard information criteria are not suitable for panels. In this case, the estimated responses of real GDP gradually return to zero and become statistically insignificant at the 10% level for $c \geq 7$.[18] Furthermore, for twin financial crises, the estimated long-run effects of 2.9% based on $A = B = 4$ are not included in the 90% confidence intervals for $c \geq 5$ when $A = 8$ and $B = 1$. Thus, we cannot reject the null of eventual economic recovery because of the estimated mean-reverting dynamics at long horizons.

This conclusion from our preferred specification remarkably differs from that of C&S. We find that their dataset and econometric method, when properly used, do not support their

[15] While C&S do not include time fixed effects, we add them for robustness.

[16] The results for the other types of crisis and sub-groups of countries can be found in Online Appendix F.

[17] The results are similar when the standard errors are clustered by country (not shown).

[18] When we choose $B = 4$ as in C&S, while maintaining $A = 8$, the results barely change from those shown in the right panels of Figure 4.

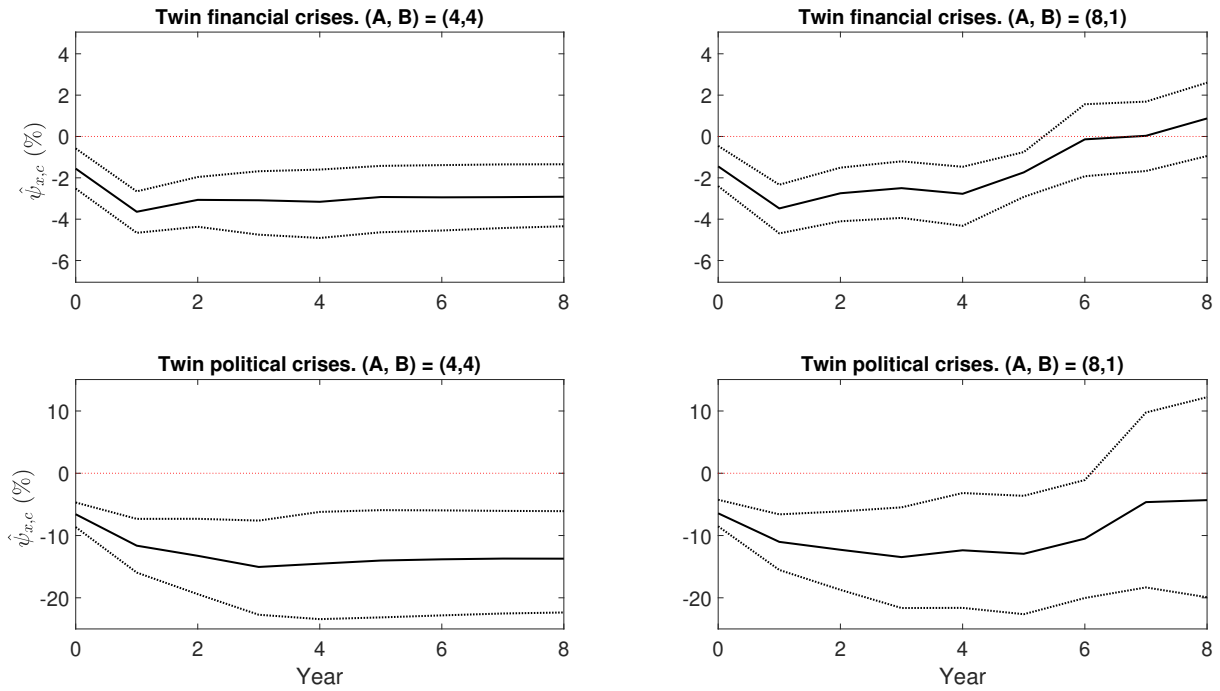

Figure 4: Effects of twin financial and political crises on real GDP

Notes: In the left panels, both A and B are four following Cerra and Saxena (2008). The right panels depict the estimates when $A = 8$ and $B = 1$. For a fair comparison, we drop observations without eight lags for both specifications. The top (bottom) panels illustrate the responses of real GDP to twin financial (political) crises, where twin financial crises denote cases in which a currency crisis and a banking crisis occur in the same year, and twin political crises imply a similar situation with stronger executive power and civil wars in the same year. Although the economy appears to not fully recover from a crisis permanently when $A = B = 4$ (the left panels), this is mostly due to the flattening effects of $\hat{\psi}_{x,c}$ for $c > A$, which is discussed in Remark 4. When we use $A = C = 8$ and $B = 1$, following our suggestion, we cannot reject the null of eventual economic recovery and the long-run neutrality of a crisis (the right panels). These results are consistent with the discussion in Section II and illustrate why we should check specifications with long enough distributed lags in ADL models. The robust standard errors are computed following Driscoll and Kraay (1998), and the 90 percent confidence intervals are shown by the dotted lines.

conclusion of permanent output loss. Instead, we interpret the long-run effects that C&S report as a statistical artifact generated by flattened impulse responses after A periods and the small value of A they select relative to C.

Efficiency and Autoregression

Next, we turn to another application to illustrate the importance of choosing B to improve the efficiency of $\hat{\psi}_{x,c}$. Specifically, we estimate how unemployment rates respond to the military news shocks constructed by Ramey and Zubairy (2018, henceforth R&Z). This

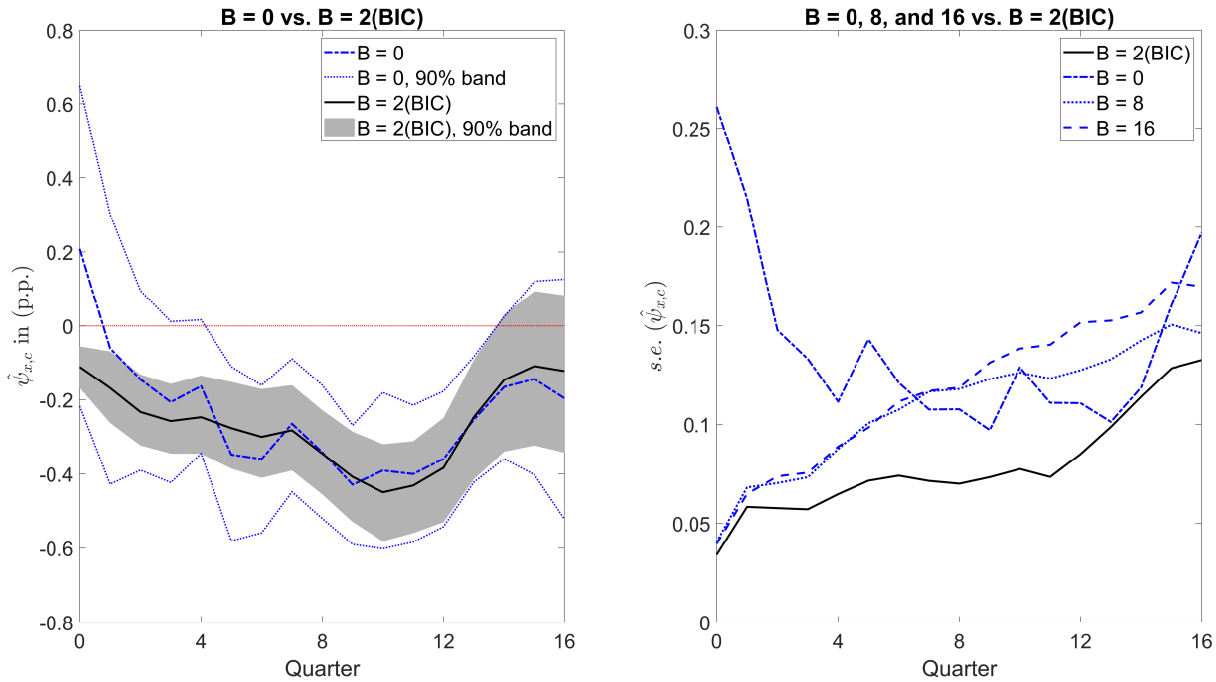

Figure 5: Effects of military news shocks on unemployment rates

Notes: We investigate the effects of military news shocks on unemployment rates using the data constructed by Ramey and Zubairy (2018). The sample period is from 1914:Q3, when World War I erupted, to 2015:Q3. In the left panel, the black solid line with the shaded area represents the estimated responses of unemployment rates and the 90 percent confidence bands when $A = 16$ and $B = 2$, which is chosen by the BIC. The input is a military news shock that the present discounted value of real government spending will increase by 10 percent of the current trend real GDP. The dash-dotted line and the dotted lines in the left panel illustrate the results for a case in which $A = 16$ and $B = 0$. The right panel plots the Newey-West HAC standard errors of $\hat{\psi}_{x,c}$ when B is either 0, 2, 8, or 16. Clearly, both too small and too large values of B increase the standard errors and render the estimator less efficient. For example, in the left panel, the standard error of $\hat{\psi}_{x,0}$ when $B = 0$ is 7.8 times as large as that under our benchmark specification with $B = 2$. Similarly, in the right panel, the standard error of the peak effects $\hat{\psi}_{x,10}$ when $B = 8$ (16) is larger than that under the case with $B = 2$ by 64% (74%). Therefore, arbitrarily selecting models either with too-long autoregressive lags or without lags is not recommended for efficient estimation.

example is provided for expositional purposes and is not directly related to the analyses in R&Z.

In the left panel in Figure 5, we have two sets of estimates. In the first case, we use $A = C = 16$ and $B = 0$. Such distributed lag models without autoregressive components are widely used in applied macroeconomic research (e.g., Alesina, Favero and Giavazzi, 2015; Coibion, Gorodnichenko and Ulate, 2018; Hsiang and Jina, 2014; Romer and Romer, 2010). In the second case, $A = C$, and B is equal to 2, which is selected by the BIC. The responses are based on a military news shock that the present discounted value of real gov-

ernment spending will increase by 10 percent of the current trend real GDP. We calculate heteroskedasticity and autocorrelation consistent (HAC) standard errors following Newey and West (1987). When $B = 0$, the unemployment rates are estimated to increase by 0.21 percentage points concurrently and gradually decrease in response to the expansionary military news shock. Furthermore, the estimates are statistically insignificant at the 10 percent level for a year after the shock due to the large standard errors. However, the variability of the estimator is substantially reduced when $B = 2$. For example, the standard error of $\hat{\psi}_{x,0}$ when $B = 2$ is only 13% of that under the model with $B = 0$. As the standard errors decrease, the estimates become statistically significant at the 10 percent level for all $c \leq 13$, including the initial decline by 0.11 percentage points.

Similarly, too large values of B also increase the standard errors and render the estimator inefficient as shown in the right panel. For example, the standard error of the peak effects $\hat{\psi}_{x,10}$ when $B = 8$ (16) is larger than that under the case with $B = 2$ by 64% (74%). Therefore, arbitrarily selecting models either with too-long autoregressive lags or without lags is not recommended for efficient estimations. Furthermore, an information criterion can substantially enhance the efficiency of $\hat{\psi}_{x,c}$ compared to the case with an arbitrarily chosen B.

In summary, we can reduce biases of the IRF estimators by setting $A \geq C$. Among asymptotically unbiased estimators ($A \geq C$), our method selects B to reduce variances. Therefore, the two applications in this section supplement our findings presented in Section II with results based on real-world data.

IV Conclusion

Our paper examines the properties of impulse response functions based on ADL models. We provide a simple guide for selecting proper lag orders to secure consistency in a generic setup and reduce the standard error of the impulse response estimator. Furthermore, we show that if an external instrument of a structural shock is available, we can consistently estimate the relative impulse responses using ADL models similar to the SVAR and LP approaches.

This paper investigates asymptotics of the estimated impulse responses given fixed A, B, and C. It is left for future research examining properties of the impulse response estimators at long horizons when these parameters increase with the sample size T.

A Proof of Proposition 2

We assume that the constant drift term g_y in Δu_t is zero without loss of generality. We have a regression $\Delta y_t = \zeta_t' \eta + v_t$, where

$$\begin{aligned} \zeta_t &= (z_t, z_{t-1}, \ldots, z_{t-A}, \Delta y_{t-1}, \Delta y_{t-2}, \ldots, \Delta y_{t-B})' \quad \text{for all } t, \\ \eta &= (\alpha_0, \ldots, \alpha_A, \beta_1, \ldots, \beta_B)'. \end{aligned}$$

By the weak law of large numbers, plim $\hat{\eta} = \bar{\eta} = (E[\zeta_t \zeta_t'])^{-1} E[\zeta_t \Delta y_t]$. Thus, we have

$$(E[\zeta_t \zeta_t']) \bar{\eta} = E[\zeta_t \Delta y_t].$$

Because $\Delta y_t = \varphi(L) x_t + \psi_e(L) e_t$, the first $A+1$ equations are given by

$$\begin{pmatrix} & 0 & 0 & \ldots & 0 \\ \sigma_z^2 I_{A+1} & \varphi_0 \rho \sigma_x \sigma_z & 0 & \ldots & 0 \\ & \vdots & \vdots & \ddots & \vdots \\ & \varphi_{A-1} \rho \sigma_x \sigma_z & \varphi_{A-2} \rho \sigma_x \sigma_z & \ldots & \varphi_{A-B} \rho \sigma_x \sigma_z \end{pmatrix} \bar{\eta} = \begin{pmatrix} \varphi_0 \\ \varphi_1 \\ \vdots \\ \varphi_A \end{pmatrix} \rho \sigma_x \sigma_z,$$

where the correlation between x_t and z_t is denoted by $\rho \neq 0$. Let θ be $\frac{Cov(x_t, z_t)}{Var(z_t)} = \rho \frac{\sigma_x}{\sigma_z}$. Furthermore, we use a bar to represent the probability limit of each estimator. Then, the above system yields the following equations:

$$\begin{aligned} \bar{\alpha}_0 &= \varphi_0 \theta, \\ \bar{\alpha}_1 + \bar{\beta}_1 \varphi_0 \theta &= \varphi_1 \theta, \\ &\vdots \\ \bar{\alpha}_A + \bar{\beta}_1 \varphi_{A-1} \theta + \ldots + \bar{\beta}_B \varphi_{A-B} \theta &= \varphi_A \theta. \end{aligned} \tag{A.1}$$

Next, we consider the estimated $\varphi(L)$, which is given by $\hat{\beta}(L)^{-1} \hat{\alpha}(L)$. In the probability limit, we have

$$\bar{\varphi}(L) = \bar{\varphi}_0 + \bar{\varphi}_1 L + \ldots = \left(1 - \bar{\beta}_1 L - \ldots - \bar{\beta}_B L^B\right)^{-1} \left(\bar{\alpha}_0 + \bar{\alpha}_1 L + \ldots + \bar{\alpha}_A L^A\right).$$

Multiplying both sides by $\left(1 - \bar{\beta}_1 L - \ldots - \bar{\beta}_B L^B\right)$ and expanding the lag polynomials gen-

erate a system of equations characterizing $(\bar{\varphi}_0, \ldots, \bar{\varphi}_A)'$:

$$\begin{aligned} \bar{\alpha}_0 &= \bar{\varphi}_0, \\ \bar{\alpha}_1 + \bar{\beta}_1 \bar{\varphi}_0 &= \bar{\varphi}_1, \\ &\vdots \\ \bar{\alpha}_A + \bar{\beta}_1 \bar{\varphi}_{A-1} + \ldots + \bar{\beta}_B \bar{\varphi}_{A-B} &= \bar{\varphi}_A. \end{aligned} \tag{A.2}$$

By comparing the first lines in (A.1) and (A.2), we obtain $\bar{\varphi}_0 = \bar{\alpha}_0 = \varphi_0 \theta$. Then, from the second lines, we obtain $\bar{\varphi}_1 - \varphi_1 \theta = \bar{\beta}_1 (\bar{\varphi}_0 - \varphi_0 \theta) = 0$. By induction, we can show that $\bar{\varphi}_c = \varphi_c \theta$ for all $c \leq A$.

As a result, $\text{plim } \hat{\psi}_{x,c} = \text{plim } (\hat{\varphi}_0 + \ldots + \hat{\varphi}_c) = \bar{\varphi}_0 + \ldots + \bar{\varphi}_c = \theta (\varphi_0 + \ldots + \varphi_c) = \theta \psi_{x,c}$ for all $c \leq A$. □

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